

Model Lab

Every prior, same slate — 2026-09-13

SERVED PRIOR

Core (KNN+SVR)

PRIORS COMPARED

10

HELD-OUT PA

35,870

Held-Out Scores

Fitted on plate appearances before 2026-08-08, scored on the 35,870 that came after (actual HR rate 2.96%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5679	1.30×	0.02866	0.13257	0.00477
KNN comps Prior is the pooled rate of the nearest comparables.	0.5876	1.48×	0.02865	0.13234	0.00507
SVR Support-vector regression on the same feature matrix.	0.5782	1.43×	0.02873	0.13405	0.00832
Random forest Forest regression; a learned similarity metric.	0.5799	1.42×	0.02874	0.13357	0.00854
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5876	1.48×	0.02865	0.13234	0.00507
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5867	1.47×	0.02866	0.13244	0.00536
Linear regression Ordinary least squares on the same features.	0.5841	1.39×	0.02870	0.13319	0.00719
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5869	1.49×	0.02871	0.13284	0.00789
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5856	1.46×	0.02868	0.13267	0.00705
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5801	1.43×	0.02879	0.13392	0.00920

Blend weights (forest variant): KNN 0.85, SVR 0.00, RF 0.15

Shrinkage K per prior: league 254, knn 254, svr 254, rf 254, core 254, ensemble 254, linear 254, logistic 254, nn 254, xgb 254 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,850 PA, actual HR rate 2.44%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5246	1.17×	0.02385	0.11534	0.00781
KNN comps Prior is the pooled rate of the nearest comparables.	0.5679	1.38×	0.02394	0.11745	0.01299
SVR Support-vector regression on the same feature matrix.	0.5504	1.17×	0.02439	0.12769	0.02506
Random forest Forest regression; a learned similarity metric.	0.5563	1.06×	0.02451	0.12522	0.02650
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5679	1.38×	0.02394	0.11745	0.01299
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5657	1.17×	0.02399	0.11820	0.01483
Linear regression Ordinary least squares on the same features.	0.5688	1.28×	0.02430	0.12419	0.02150
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5691	1.17×	0.02436	0.12109	0.02376
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5722	0.96×	0.02409	0.11926	0.01945
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5653	0.96×	0.02498	0.12775	0.03378

Slate Agreement

Flat league · 3/6 shared 2/6 hit

Luis García Jr.
Francisco Lindor
Kyle Schwarber
Ben Rice ✓
Junior Caminero
Eugenio Suárez ✓

SVR · 5/6 shared 1/6 hit

Kyle Schwarber
Luis García Jr.
Ben Rice ✓
Juan Soto
James Wood
Junior Caminero

Core (KNN+SVR) · 6/6 shared 1/6 hit

Yohandy Morales
Juan Soto
Kyle Schwarber
Ben Rice ✓
James Wood
Luis García Jr.

Linear regression · 4/6 shared 0/6 hit

Emmanuel Rodriguez
Lazaro Montes
Yohandy Morales
Kyle Schwarber
Juan Soto
Luis García Jr.

Neural net (torch) · 6/6 shared 1/6 hit

KNN comps · 6/6 shared 1/6 hit

16.8%	Yohandy Morales	19.4%
16.8%	Juan Soto	19.2%
16.4%	Kyle Schwarber	19.1%
16.3%	Ben Rice ✓	18.9%
16.1%	James Wood	18.7%
16.1%	Luis García Jr.	18.5%

Random forest · 5/6 shared 1/6 hit

20.4%	Emmanuel Rodriguez	21.0%
20.3%	Yohandy Morales	20.6%
20.1%	Kyle Schwarber	19.9%
19.8%	Ben Rice ✓	19.7%
19.5%	Juan Soto	19.6%
19.4%	Luis García Jr.	19.6%

Forest blend · 6/6 shared 1/6 hit

19.4%	Yohandy Morales	19.6%
19.2%	Juan Soto	19.3%
19.1%	Kyle Schwarber	19.2%
18.9%	Ben Rice ✓	19.0%
18.7%	James Wood	18.7%
18.5%	Luis García Jr.	18.7%

Logistic regression · 4/6 shared 0/6 hit

25.5%	Emmanuel Rodriguez	27.2%
21.2%	Lazaro Montes	23.7%
19.9%	Kyle Schwarber	22.2%
19.8%	Yohandy Morales	21.3%
19.2%	Luis García Jr.	20.0%
19.0%	Juan Soto	19.8%

James Wood

Ben Rice ✓

1/6 hit

19.5%

Hitter by Hitter

[illegible]

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
8. Emmanuel Rodriguez slate Minnesota Twins · vs RHP · 1 HR / 4 PA	11.1%	17.9%	9.8%	21.0%	17.9%	18.4%	25.5%	27.2%	18.1%
9. Cal Raleigh Seattle Mariners · vs RHP · 21 HR / 477 PA	14.6%	17.4%	16.1%	16.7%	17.4%	17.3%	17.9%	18.9%	17.6%
10. Abimelec Ortiz Washington Nationals · vs RHP · 5 HR / 117 PA	14.4%	17.4%	16.3%	17.5%	17.4%	17.4%	17.9%	17.3%	18.2%
11. Francisco Lindor New York Mets · vs RHP · 20 HR / 402 PA	16.8%	17.3%	17.6%	17.3%	17.3%	17.3%	17.5%	17.6%	17.3%
12. Junior Caminero Tampa Bay Rays · vs RHP · 40 HR / 647 PA	16.1%	17.2%	19.4%	17.4%	17.2%	17.2%	17.3%	17.9%	17.8%
13. Eugenio Suárez Cincinnati Reds · vs RHP · 23 HR / 455 PA	16.1%	17.2%	17.7%	17.6%	17.2%	17.3%	17.4%	17.5%	17.3%
14. Yordan Alvarez Houston Astros · vs RHP · 38 HR / 636 PA	14.5%	17.1%	18.3%	18.1%	17.1%	17.3%	17.6%	19.3%	18.4%
15. Ronald Acuña Jr. Atlanta Braves · vs RHP · 17 HR / 421 PA	15.4%	16.9%	16.7%	16.4%	16.9%	16.8%	16.4%	16.6%	16.7%
16. CJ Abrams Washington Nationals · vs RHP · 30 HR / 609 PA	14.6%	16.9%	16.7%	16.6%	16.9%	16.8%	15.9%	16.3%	16.1%
17. Rafael Devers San Francisco Giants · vs RHP · 37 HR / 633 PA	14.7%	16.8%	17.1%	17.0%	16.8%	16.9%	16.5%	17.3%	16.9%
18. Lazaro Montes Seattle Mariners · vs RHP · 3 HR / 27 PA	11.7%	16.7%	12.4%	18.6%	16.7%	17.0%	21.2%	23.7%	18.2%
19. Aaron Judge New York Yankees · vs RHP · 17 HR / 276 PA	12.2%	16.4%	17.6%	17.2%	16.4%	16.5%	17.5%	18.8%	17.9%
20. Sal Stewart Cincinnati Reds · vs RHP · 31 HR / 641 PA	14.8%	16.1%	16.8%	16.7%	16.1%	16.2%	16.1%	15.9%	16.1%
21. Coby Mayo Baltimore Orioles · vs LHP · 23 HR / 402 PA	13.4%	16.0%	16.5%	16.5%	16.0%	16.1%	16.1%	17.5%	16.5%
22. Oneil Cruz	13.4%	16.0%	16.5%	15.7%	16.0%	15.9%	15.5%	15.4%	15.8%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
Pittsburgh Pirates · vs RHP · 18 HR / 375 PA									
23. Bryce Harper Philadelphia Phillies · vs RHP · 28 HR / 634 PA	14.3%	15.8%	16.0%	16.2%	15.8%	15.9%	15.6%	15.7%	15.9%
24. Spencer Jones New York Yankees · vs RHP · 9 HR / 243 PA	13.4%	15.8%	15.3%	14.8%	15.8%	15.6%	13.9%	13.9%	14.4%
25. Zach Neto Los Angeles Angels · vs RHP · 25 HR / 639 PA	13.7%	15.7%	15.8%	15.4%	15.7%	15.6%	15.0%	15.7%	15.1%
26. Kyle Stowers Miami Marlins · vs RHP · 19 HR / 468 PA	12.7%	15.6%	15.6%	15.1%	15.6%	15.6%	14.4%	15.5%	15.4%
27. Jackson Chourio Milwaukee Brewers · vs LHP · 23 HR / 520 PA	14.8%	15.6%	16.4%	16.7%	15.6%	15.8%	16.0%	16.2%	16.1%
28. Pete Alonso Baltimore Orioles · vs LHP · 35 HR / 647 PA	12.8%	15.6%	16.0%	15.5%	15.6%	15.6%	15.2%	16.0%	15.5%
29. Tyler Stephenson Cincinnati Reds · vs RHP · 15 HR / 427 PA	13.7%	15.5%	13.9%	14.9%	15.5%	15.4%	14.7%	14.3%	14.3%
30. Corey Seager Texas Rangers · vs RHP · 17 HR / 380 PA	12.1%	15.5%	14.9%	15.3%	15.5%	15.4%	14.9%	15.1%	15.3%
31. Elly De La Cruz Cincinnati Reds · vs RHP · 26 HR / 576 PA	14.0%	15.4%	16.1%	15.0%	15.4%	15.3%	15.0%	15.2%	15.2%
32. Jake Bauers Milwaukee Brewers · vs LHP · 25 HR / 532 PA	12.9%	15.3%	15.2%	15.8%	15.3%	15.4%	14.6%	15.0%	15.1%
33. Rafael Flores Jr. Pittsburgh Pirates · vs RHP · 8 HR / 123 PA	12.3%	15.3%	16.6%	15.3%	15.3%	15.3%	16.2%	15.8%	16.4%
34. Zack Gelof Athletics · vs LHP · 17 HR / 381 PA	13.5%	15.2%	15.5%	14.8%	15.2%	15.2%	15.3%	15.2%	15.4%
35. Gary Sánchez Milwaukee Brewers · vs LHP · 11 HR / 254 PA	14.1%	15.2%	15.9%	16.0%	15.2%	15.3%	15.3%	14.1%	15.5%
36. Andrés Chaparro Washington Nationals · vs RHP · 10 HR / 197 PA	13.9%	15.2%	15.9%	15.4%	15.2%	15.2%	15.0%	15.1%	15.7%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
37. Drake Baldwin Atlanta Braves · vs RHP · 23 HR / 559 PA	13.5%	15.2%	15.3%	15.6%	15.2%	15.2%	15.0%	15.0%	15.0%
38. Austin Riley Atlanta Braves · vs RHP · 17 HR / 582 PA	13.3%	15.1%	15.0%	14.5%	15.1%	15.0%	14.4%	14.9%	14.9%
39. Pete Crow-Armstrong Chicago Cubs · vs LHP · 41 HR / 668 PA	13.3%	15.1%	15.6%	15.5%	15.1%	15.2%	14.8%	16.0%	15.4%
40. Jeremy Peña Houston Astros · vs RHP · 17 HR / 428 PA	14.6%	15.1%	15.2%	14.9%	15.1%	15.1%	14.5%	14.5%	14.5%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.